

UDAY RAM V

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CAREER SUMMARY

A **Data Science** and **programming enthusiast**, aiming for **Automation** and **Optimization** of recurrent process possible to see the world working on something worthwhile, with a strong foundation in **Software development** with proficiency in **Python, C, C++, Java, SQL, NoSQL**. Experienced in building **high dimensional statistical machine learning models**, **data analysis**, **data preprocessing**, **data visualizations**, **automation applications**, **web applications**.

TECHNICAL SKILLS

Programming Languages	Python, Java, C, C++, R, JavaScript, Shell Script.
Databases	MySQL, PL/SQL, Neo4j (Graph SQL), NoSQL, Postgres, Snowflake (<i>SnowPro certified</i>).
Machine Learning and Deep Learning	SAS programming, Pandas, Numpy, Scikit-Learn, Keras, TensorFlow, PyTorch, LLM, NLP, CNN, MATLAB.
Frameworks	Flask, Django, Pytest, PyUnit.
Cloud Databases	GCP Clusters, GCP BigQuery, GCP DataProc, AWS Data Lake, AWS Lambda, AWS Key Vault, Azure Data Lake, Azure Key Vault.
Big Data Technologies	Hive, Hadoop, Spark, Kafka, Data Structures.
Software Tools	Tableau, Open Refine, RapidMiner, Power BI, Open CV, Microsoft Office, Googles Docs sheets, Bitbucket, Confluence, Jira, Docker, Jenkins, IntelliJ Idea, PyCharm, VS Code, Git, Airflow, Vision, Gray Log.

EDUCATION

University of North Texas, Denton, Texas

Aug 2022 – Dec 2023

Master of Science in Data Science

GPA: 3.8

Relevant Coursework: **Applied Machine Learning, Statistical models, Data Modeling, Data Analysis, Data Mining, Data Visualization.**

SRM Institute of Science and Technology, India

June 2015 – July 2019

Bachelor of Technology in Computer Science and Engineering

GPA: 3.5

Relevant Coursework: **Python Programming, Advanced Data Structures, Machine Learning Algorithm Implementation.**

PROFESSIONAL EXPERIENCE

ML Engineer at **Ford**, TX, USA

Dec 2022 – Present

Technologies: **Python, Data Analysis, Machine Learning Models, Deep Learning, Tableau, Deployment**

- Worked on the datasets with **Data Pre-processing, Data preparation and Data transformations** using **Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, Stats**.
- Engineered features from structured and unstructured data sources, leveraging **domain knowledge** and techniques using **text vectorization, image augmentation, and time series decomposition**.
- Conducted comprehensive data analysis and preprocessing techniques, including **missing value imputation, outlier detection, and feature scaling**, using **Pandas** and **NumPy**.
- Developed end-to-end machine learning solutions, from **data preprocessing** and **feature engineering** to model training and evaluation with **classification, regression, clustering, reinforcement learning, convolutional neural networks (CNN), natural language processing (NLP)**, using Python **scikit-learn, Keras, TensorFlow, and PyTorch**.
- Deployed machine learning models into variety of environments using **Docker** with **Kubernetes** platforms for scalability and reproducibility, to integrate models with **CI/CD workflows, validation and deployment** using **Jenkins**.

ML Engineer at NielsenIQ Company Pvt Ltd, India

June 2019 – July 2022

Technologies: **Python, SQL, Neo4J Graph SQL, Machine learning, Snowflake, Postgres.**

- Worked in **Python** utilizing libraries such as **scikit-learn, TensorFlow, and Keras** to design, build, and deploy predictive models on large datasets.
- Developed a **churn prediction model** using Python's **scikit-learn, PyTorch** library by training various machine learning algorithms including **logistic regression, random forest, and gradient boosting** on a large dataset to identify customers at risk of churn and deployed the model using **Flask** for real-time predictions.
- Evaluated **model performance** using appropriate metrics and **cross-validation techniques, optimizing hyperparameters, grid search, random search, Bayesian optimization and accuracy, precision, recall, F1-score, ROC-AUC.**
- Performing **data transformations** based on **Performance and efficiency** of huge data using **pandas, NumPy, pivot, unpivot operations** in **Snowflake SQL and PostgreSQL.**
- Worked with Fine-tuned pre-trained Language Models (LLMs) - **BERT, GPT, and RoBERTa** using framework **OpenAI GPT.**
- Developed and deployed **performant statistical models** for various applications, including **time series forecasting and real-time data analysis.**
- Utilized techniques such as Integrated Gradients, **LIME** (Local Interpretable Model-Agnostic Explanations), and **SHAP** (SHapley Additive exPlanations) to interpret **BERT** model predictions and provide explanations for decision-making processes.
- Improved and applied **predictive modeling techniques and machine learning algorithms** to large datasets, achieving actionable insights and predictive accuracy.
- Hands-on experience with **unstructured data** processing, leveraging **Natural Language Processing (NLP)** techniques for **text data analysis.**
- Explored **Generative AI models**, including Variational autoencoders (VAEs) and Generative adversarial networks (GANs), for generating synthetic data and enhancing enterprise applications.
- Familiar in Setting up and configuring **CI/CD pipelines** for Python applications using **Terraform, Jenkins, Docker image and Kubernetes.**
- Created **DAG's and Rest APIs** for triggering dataset ingestion, importing and fetching using **python flask** application in **AIRFLOW and VISION.**
- **Set up, Built and Deployed** in various environments with **Automated SHELL Scripts** using **Jenkins.**
- Owned and developed an **Automation Flask application** from the scratch to seamlessly implement **RESTful APIs**, specializing in **handling large-scale data operations** that include data collection, preparation, transformation, and ingestion using **python** in **Snowflake, Postgres.**
- Developed and Managed **Automated Testing Framework** for all the Rest APIs and basic JWT token generation features.

Data Analyst Intern at NielsenIQ Company Pvt Ltd, India

Feb 2019 – May 2019

Technologies: **Python, Data Analysis, Machine Learning Models, Automation Framework**

- Worked on the dataset with **Data Pre-processing, Data preparation and Data transformations** using **Pandas, NumPy, Scikit-learn, Seaborn, Matplotlib, Stats.**
- Implemented and Applied various **Machine Learning Algorithms/Models** that include **Linear and Logistic Regressions, SVM, K-NN, NLP** to find the pattern of the data prepared.
- Used **Data Visualizations** for presenting the data patterns found using various applied models on data.
- **Developed an Automation Testing Framework** in Python for testing huge number of APIs, fundamental operations of an entire application.

PROJECT EXPERIENCE

Product Review Aspect Ranking

- Aim – to get a consolidated and meaningful opinion of all the reviews of a product.
- Implemented a model using **Text Classification, Semantic Analysis, Tokenization**, and other **NLP Algorithms** to find out the aspect of reviews of a product using **Python, Scikit-Learn, Pandas, NumPy**.
- Applied the model and performed the operations on a set of reviews of few products for classification and found a success rate of **87%** of the model.

Image Recognition

- Aim – to identify an image based on the set of trained images of animals.
- Performed Image preprocessing on the data for converting all the images to Uni colored images with a proper outline using **Python, Scikit-Learn**
- Implemented **K-Means algorithm** for the preprocessed dataset and resulted in **89% accuracy** rate of the model.

Exploratory Data Analysis on GDP Dataset

- Aim – to perform analysis and identify various patterns involved in the dataset.
- Exploratory **Data Analysis** on GDP dataset from World Development Indicators: Conducted thorough **data exploration** on the dataset, revealing insights and presenting them with various visualizations using **Python**.

Data Analysis in Google Cloud Platform (GCP) and Tableau

- Created **VM instances** in **GCP Clusters**, to upload and use the dataset from **BigQuery** for data analysis.
- Initiated **Spark, HIVE** instances and performed few queries for data analysis on the dataset imported using **Spark SQL** and **Hive SQL**.
- Also Preprocessed the data in **OpenRefine** tool using diverse options and visualized the data in various plots and graphs in **Tableau**.

INTERESTS

Programming | Applied Machine Learning | Automation | Data Analysis | Data Engineering

ACHIEVEMENTS

- Successfully built **IRIS Recognition model** using **Daugman's** algorithm with **80 percent accuracy**.
- **Improved business logic and performance** that saved the budget of the organization by improving ingestion process from a **12-hour process to 2:40 minutes** in handling huge data by implementing an **automated PL/SQL and python**, that chunks, transforms and ingests data.
- Developed **end-to-end Automation of data ingestion process from scratch**, that gets that data from ADLs and Postgres, performs transformation processes using Pandas, ingests to Snowflake to perform pivot operations and then uploads to ADLs.